IAT New Test 4

Question paper, answer key
and worked solutions

> **MANUAL REVIEW DRAFT**
> 2027 preparation using the verified 2026 pattern and stored 2026 syllabus.
> The 2027 brochure has not been verified. Not approved for student publication.

60	180	240
single-correct questions	minutes	maximum marks

Subject	Question numbers	Marks
Biology	1-15	60
Chemistry	16-30	60
Mathematics	31-45	60
Physics	46-60	60

Instructions

1. There are 60 single-correct questions: Biology 1–15, Chemistry 16–30, Mathematics 31–45 and Physics 46–60.
2. Each question has exactly one correct option. Correct: +4; incorrect: -1; unanswered: 0. All questions are available to attempt; there is no best-three-subject scoring rule.
3. Use 180 minutes. An on-screen non-programmable scientific calculator is part of the 2026 examination conditions; personal calculators are not allowed under that scheme.
4. Treat stated data and idealisations as exact unless an approximation is explicitly supplied. Log denotes base 10 and ln denotes the natural logarithm.
5. The review PDF contains the question paper first, followed by a separate answer key and worked solutions. Keep the solutions hidden during a timed attempt.

Fixed authored order. Figures and IDs must remain linked during review.
Revised 2026-10-31 · IAT27_TEST_04 · Original AI-assisted content.

QUESTION PAPER / B

Biology

Questions 1-15 · 15 questions · 60 marks

Q1

+4 / -1

A woody plant bears pollen and ovules on separate cones; its ovules are exposed at pollination, although a seed coat develops later. Which observation distinguishes it from an angiosperm?

A) Ovules are not enclosed in an ovary before fertilisation

B) It produces pollen grains

C) Its diploid zygote develops into an embryo

D) It forms vascular tissue

Q2

+4 / -1

The figure depicts a floral section: sepals, petals and stamens arise above the ovary; the ovary has two chambers. If the flower is turned upside down for display, which classification remains valid?

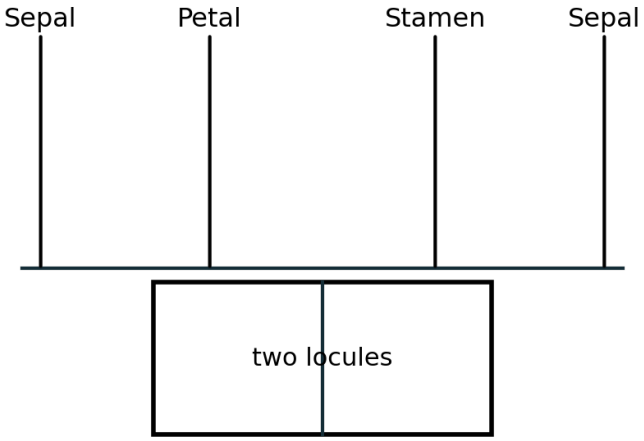


Figure for Q2

A) Superior, unilocular ovary

B) Inferior, bilocular ovary

C) Superior, bilocular ovary

D) Inferior, unilocular ovary

Q3

+4 / -1

Complete hydrolysis of one unbranched peptide releases 18 amino-acid molecules. How many water molecules were consumed if no cyclic bond was present?

A) 18

B) 16

C) 17

D) 19

Q4

+4 / -1

A tissue consumes 20 mL O_2 and releases 16 mL CO_2 in a short closed-chamber trial. What is its measured respiratory quotient, and which pure-substrate value is it closer to: glucose 1.0 or a typical fat 0.7?

A) 1.25, glucose

B) 1.25, fat

C) 0.8, glucose

D) 0.8, fat

Q5

+4 / -1

After a booster dose of the same antigen, serum antibody rises sooner and higher than after the first exposure. Which cellular population most directly explains the difference?

A) Long-lived memory B cells selected during the first exposure

B) Mature erythrocytes that bind antigen

C) Platelets that secrete antigen-specific IgG

D) Neutrophils converting into plasma cells

Q6

+4 / -1

In an angiosperm, a fertilised ovule develops normally but the surrounding ovary wall is experimentally removed immediately after fertilisation. Which structure can still arise directly from the ovule?

A) Pericarp

B) Seed with embryo and seed coat

C) Fruit wall

D) Receptacle

Q7

+4 / -1

An allele begins at frequency 0.10 in eight large replicate populations. After many generations under the same insecticide, it rises above 0.80 in all eight; without insecticide, frequencies fluctuate around 0.10. What is the strongest inference?

A) Genetic drift must cause the consistent directional rise

B) Insecticide directly edits the allele in every individual

C) The data support selection favouring the allele under insecticide

D) The allele is necessarily dominant

Q8

+4 / -1

The figure shows a simplified malaria cycle with stages P in mosquito salivary glands, Q in human liver cells, R in human red cells and S as gametocytes in human blood. Which stage enters a human with the mosquito bite?

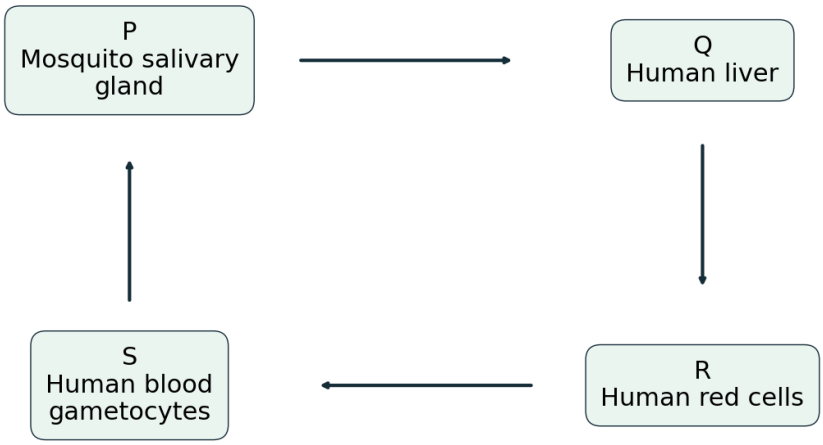


Figure for Q8

A) P, sporozoites

B) Q, liver schizonts

C) R, merozoites in erythrocytes

D) S, gametocytes

Q9

+4 / -1

A plasmid carries antibiotic-resistance genes ampR and tetR; foreign DNA interrupts tetR. Colonies grow on ampicillin but fail to grow on tetracycline. What does this pattern support?

A) No plasmid entered the cells

B) A recombinant plasmid disrupted tetR

C) Both resistance genes remain intact

D) The insert disrupted ampR

Q10

+4 / -1

Producers capture 80,000 kJ; approximately 10% of available energy transfers at each trophic step. What energy reaches tertiary consumers through producers→herbivores→secondary→tertiary consumers?

A) 8,000 kJ

B) 800 kJ

C) 80 kJ

D) 8 kJ

Q11

+4 / -1

A dihybrid testcross yields four phenotype classes of 98, 102, 101 and 99 offspring. Which conclusion best fits these counts?

A) Complete linkage of the two loci

B) A 9:3:3:1 ratio from two heterozygotes

C) One locus is lethal in homozygotes

D) Approximate independent assortment of the two loci

Q12

+4 / -1

Two unaffected parents have an affected child with a fully penetrant autosomal recessive condition. Their next child is unaffected. What is the probability that this unaffected child is a carrier?

A) 2/3

B) 1/2

C) 1/4

D) 3/4

Q13

+4 / -1

Heart rate rises from 60 to 90 beats/min while stroke volume falls from 80 to 60 mL/beat. What is the change in cardiac output?

A) Decrease by 0.6 L/min

B) Increase by 0.6 L/min

C) Increase by 2.4 L/min

D) No change

Q14

+4 / -1

An aquaporin increases water permeability of a membrane without consuming ATP. Which observation is compatible with this channel?

A) Water always moves from higher solute concentration to lower

B) Aquaporin pumps solutes against their gradient

C) Net water movement follows a water-potential gradient

D) Transport stops whenever ATP synthesis stops

Q15

+4 / -1

For one region S=cA^z. Species counts are 20 and 40 at areas A and 4A, respectively. What is z?

A) 1

B) 2

C) 1/4

D) 1/2

QUESTION PAPER / C

Chemistry

Questions 16-30 · 15 questions · 60 marks

Q16

+4 / -1

In the displayed structure $CH_2{=}CH{-}C{\equiv}N$, how many σ and π bonds are present, including C–H bonds?



Figure for Q16

A) 6 σ and 3 π

B) 5 σ and 3 π

C) 6 σ and 2 π

D) 7 σ and 2 π

Q17

+4 / -1

A complex ion $[M(NH_3)_4Cl_2]^+$ contains neutral NH_3 and chloride ligands. What is M's oxidation state and its coordination number?

A) +1, 6

B) +3, 6

C) +2, 4

D) +3, 4

Q18

+4 / -1

In one valid Lewis resonance contributor of NO_3^- with one N=O and two N–O single bonds, what are the formal charges on N and each singly bonded O?

A) N 0; each O −1

B) N +1; each O 0

C) N −1; each O +1

D) N +1; each O −1

Q19

+4 / -1

A hydrogen atom in an excited level emits exactly two photons in succession and ends at n=1. Their wavenumbers are R(1/4−1/16) and R(1−1/4), in that order. Which sequence of levels produced them?

A) 4→2→1

B) 3→2→1

C) 4→3→1

D) 5→2→1

Q20

+4 / -1

A secondary carbocation can undergo a 1,2-hydride shift to a tertiary carbocation without changing total molecular formula. Which statement best predicts the change?

A) It necessarily forms a less stable primary ion

B) The rearrangement can be favourable because hyperconjugation stabilises the tertiary centre

C) Hydride transfer adds a hydrogen atom from solvent

D) A hydride shift changes the net positive charge to negative

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Q21

+4 / -1

Which product pair forms when CH_3COCl undergoes complete hydrolysis in water?

A) CH_3CHO and Cl_2

B) CH_3OH and HCl

C) CH_3COOH and HCl

D) CH_3COCH_3 and H_2O

Q22

+4 / -1

An amine reacts with benzenesulfonyl chloride; its sulfonamide product dissolves in aqueous NaOH but reprecipitates upon acidification. Which class of starting amine is indicated?

A) Primary amine

B) Secondary amine

C) Tertiary amine

D) Quaternary ammonium salt

Q23

+4 / -1

A neutral-side-chain amino acid has $pK_a(COOH)=2$ and $pK_a(NH_3^+)=9$. At pH 6, which form predominates?

A) $H_3N^+–CHR–COOH$, net +1

B) $H_3N^+–CHR–COO^-$, net 0

C) $H_2N–CHR–COO^-$, net −1

D) $H_2N–CHR–COOH$, net 0

Q24

+4 / -1

The figure plots ln k against 1/T for reactions X and Y with straight lines of slopes −4000 K and −8000 K respectively. Assuming the same units and Arrhenius behaviour, which has the greater activation energy?

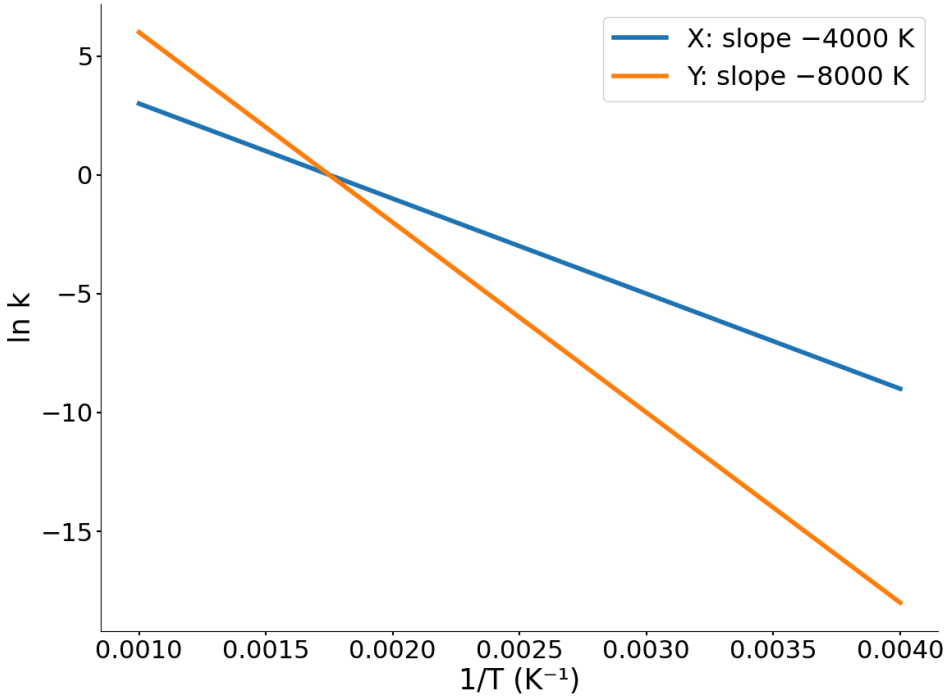


Figure for Q24

A) X, by a factor of two

B) X and Y are equal

C) Slopes alone cannot compare activation energies

D) Y, by a factor of two

Q25

+4 / -1

For AgCl, Ksp=10^{-10}. What is its approximate molar solubility in 0.010 M NaCl, assuming the added chloride dominates dissolved AgCl?

A) 10^{-5} M

B) 10^{-6} M

C) 10^{-8} M

D) 10^{-12} M

Q26

+4 / -1

Chlorine reacts with cold dilute NaOH to produce Cl^- and ClO^-. Which electron-accounting statement is consistent with the balanced reaction?

A) Both chlorine atoms are reduced from 0 to −1

B) One chlorine atom loses one electron while another gains one electron

C) Chlorine's average product oxidation state is +1

D) Oxygen is oxidised from −2 to 0

Q27

+4 / -1

Equal masses of the same liquid at 20 °C and 80 °C are mixed in an insulated vessel with negligible heat capacity and no phase change. What is the final temperature?

A) 20 °C

B) 80 °C

C) 60 °C

D) 50 °C

Q28

+4 / -1

An ideal liquid mixture contains mole fraction $x_A=0.25$. Pure vapour pressures at this temperature are $p_A^\circ=80$ kPa and $p_B^\circ=40$ kPa. What is A's fraction of the total vapour pressure?

A) 1/2

B) 1/3

C) 2/3

D) 2/5

Q29

+4 / -1

Nitration of toluene gives mostly ortho and para products. What change to the ring substituent would instead favour meta nitration?

A) Replace CH_3 with NO_2

B) Replace CH_3 with OCH_3

C) Replace CH_3 with NH_2

D) Replace CH_3 with Cl

Q30

+4 / -1

At room temperature, benzyl chloride undergoes nucleophilic substitution readily, while chlorobenzene resists ordinary SN1/SN2 substitution. Which distinction best explains this?

A) Benzyl chloride lacks a C–Cl bond

B) Chlorobenzene forms a more stable phenyl cation than a benzyl cation

C) Chlorobenzene's aryl C–Cl bond has partial double-bond character and its carbon is sp^2

D) A benzyl carbon cannot undergo backside attack

QUESTION PAPER / M

Mathematics

Questions 31-45 · 15 questions · 60 marks

Q31

+4 / -1

A hyperbola centred at the origin has transverse vertices (±3,0) and foci (±5,0). Which pair of lines are its asymptotes?

A) $y=\pm(4/3)x$

B) $y=\pm(3/4)x$

C) $y=\pm(5/3)x$

D) $y=\pm(4/5)x$

Q32

+4 / -1

Every line $(k+1)x+(2k-1)y=3k+2$ passes through one fixed point. What is that point?

A) (1,1)

B) (7/3,1/3)

C) (2,0)

D) (1,2)

Q33

+4 / -1

Evaluate $\int$ from 0 to 1 of $(2x+3)/[(x+1)(x+2)]\,dx$.

A) ln 2

B) ln(3/2)

C) ln 3

D) ln 6

Q34

+4 / -1

What is the area enclosed by $y=x^2$ and $y=2x$ between their intersections?

A) 1/3

B) 2/3

C) 8/3

D) 4/3

Q35

+4 / -1

For $x^2+xy+y^2=7$, the point (1,2) lies on the curve. What is the slope of its tangent there?

A) $-4/5$

B) $-5/4$

C) 4/5

D) 0

Q36

+4 / -1

On integers define aRb when 4 divides (a−b). If the domain is restricted to odd integers, how many equivalence classes remain?

A) 1

B) 4

C) 2

D) Infinitely many

Q37

+4 / -1

Let A=[[1,1],[0,1]], B=[[1,0],[1,1]], and AX=B. What is tr(X)?

A) 3

B) 2

C) 0

D) 1

Q38

+4 / -1

A fair coin is tossed four times. Given that at least one head occurs, what is the probability of exactly two heads?

A) 3/8

B) 2/5

C) 1/2

D) 6/16

Q39

+4 / -1

For acute α and β, tan α=1/2 and tan β=1/3. What is cos(α+β)?

A) 1/√10

B) 3/√50

C) 1/√2

D) 2/√5

Q40

+4 / -1

What is the foot of the perpendicular from P(2,−3,5) to the xy-plane?

A) (0,−3,5)

B) (2,0,5)

C) (0,0,5)

D) (2,−3,0)

Q41

+4 / -1

Vectors a=(1,2,0) and b=(2,1,0) form two sides of a parallelogram. Which vector gives its oriented area a×b?

A) (0,0,−3)

B) (0,0,3)

C) (3,3,0)

D) (0,0,5)

Q42

+4 / -1

Let ω be a non-real cube root of unity. What is $|1-\omega|^2$?

A) 1

B) 2

C) 3

D) 4

Q43

+4 / -1

For which real t does $x^2-tx+4=0$ have two distinct positive real roots?

A) $t>0$

B) $t\geq 4$

C) $t<-4$

D) $t>4$

Q44

+4 / -1

For positive integer p in {1,2,4,10}, which p makes the constant-term coefficient of $(x^2+x^{(-p)})^6$ largest?

A) 2

B) 1

C) 4

D) 10

Q45

+4 / -1

The infinite geometric series $1+r+r^2+\cdots$ has a finite sum for which condition on real r?

A) $r\leq 1$

B) $|r|<1$

C) $|r|\leq 1$

D) $r>-1$

Physics

Questions 46-60 · 15 questions · 60 marks

Q46

+4 / -1

Voltage and current are measured independently with maximum percentage uncertainties 2% and 1%. For R=V/I, what is the maximum first-order percentage uncertainty in R?

A) 3%

B) 2%

C) 1%

D) 0.02%

Q47

+4 / -1

A 5 kg block rests on a horizontal rough floor with μs=0.4. Taking g=10 m/s², what horizontal force first reaches the limiting static friction?

A) 10 N

B) 20 N

C) 50 N

D) 2 N

Q48

+4 / -1

Two wires of equal length L and cross-section A, with Young moduli Y and 2Y, are joined end to end and pulled by force F. What is their combined extension?

A) FL/(2AY)

B) FL/(AY)

C) 2FL/(AY)

D) 3FL/(2AY)

Q49

+4 / -1

In the figure, a metal sphere just fits through a circular hole in a metal plate at 20 °C. Both are heated equally; the sphere's expansion coefficient is twice the plate's. What happens for a small temperature rise?

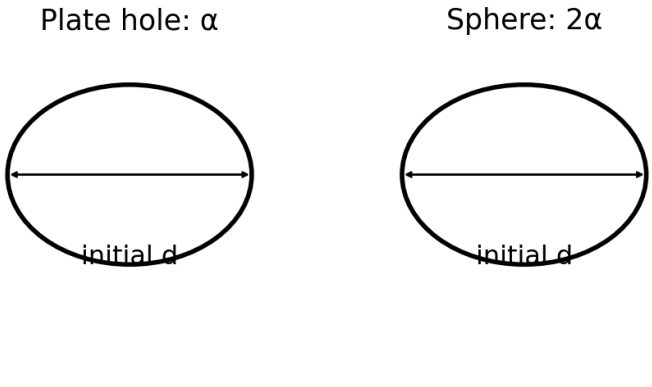


Figure for Q49

A) The hole contracts while the sphere grows

B) Both sizes grow at exactly the same fractional rate

C) The sphere can no longer pass through the hole

D) The hole grows twice as fast fractionally

Q50

+4 / -1

A rigid insulated container holds one mole of ideal diatomic gas with vibrational modes inactive. A paddle does $5RT_0/2$ work on gas initially at T_0, with no heat transfer. What is final temperature?

A) $3T_0/2$

B) $5T_0/2$

C) $3T_0$

D) $2T_0$

Q51

+4 / -1

A small-amplitude pendulum is lengthened from L to 1.44L while g stays fixed. What is the ratio of new to old period?

A) 1.44

B) 1.20

C) 0.83

D) 2.07

Q52

+4 / -1

An unknown tone gives 4 beats/s with 256 Hz and 8 beats/s with 260 Hz. What is its frequency?

A) 252 Hz

B) 264 Hz

C) 260 Hz

D) 256 Hz

Q53

+4 / -1

In 0<x<2 m a uniform electric field is +3 V/m along x. If V(0)=10 V, what is V(2)?

A) 16 V

B) 10 V

C) 7 V

D) 4 V

Q54

+4 / -1

A wire with carrier density n and drift speed v has current density J. A second wire has carrier density n/2 and drift speed 2v, with the same charge magnitude. What is its current density?

A) J/4

B) J

C) 2J

D) 4J

Q55

+4 / -1

An ideal inductor stores energy U at current I. Its current is first doubled, then the inductance is halved while current remains doubled. What is the final energy in units of U?

A) U/2

B) U

C) 2U

D) 4U

Q56

+4 / -1

The figure compares current amplitude versus driving frequency for series RLC circuits P and Q with the same L and C. P has a taller, narrower resonance peak than Q. Which is consistent?

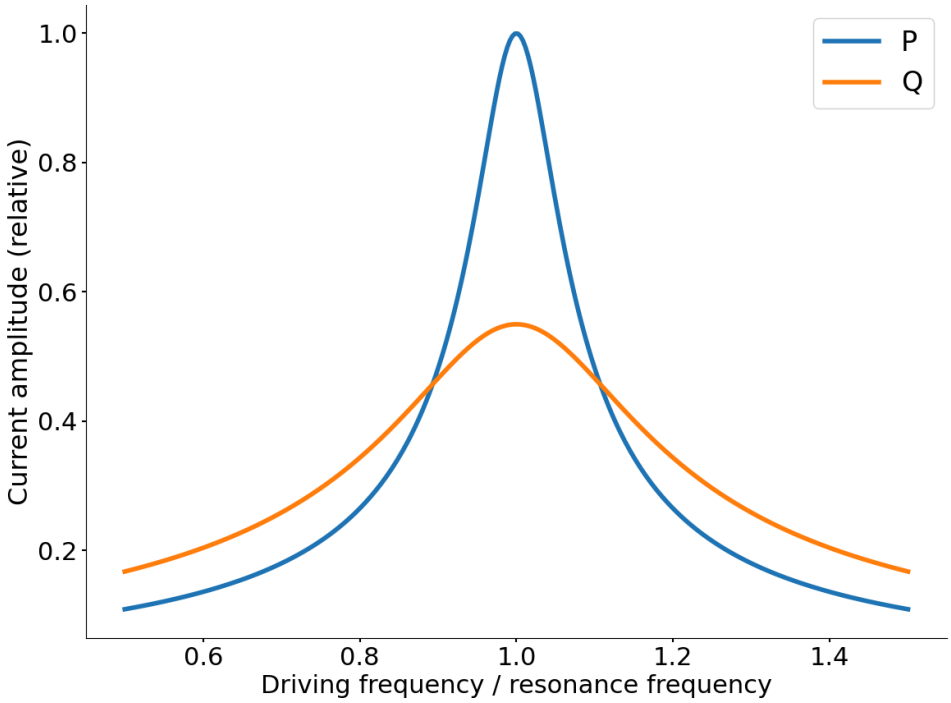


Figure for Q56

A) P has greater resistance and greater bandwidth

B) P has greater resistance but smaller bandwidth

C) Both necessarily have different resonance frequencies

D) P has smaller resistance and smaller bandwidth

Q57

+4 / -1

A prism has angle 60° and minimum deviation 30° in air. What is its refractive index?

A) √2

B) √3

C) 2

D) 1/√2

Q58

+4 / -1

A film of refractive index n lies between air and a glass substrate of index greater than n. At normal incidence, what condition gives constructive interference of the two principal reflected rays, neglecting additional reflections?

A) $2nt=(m+1/2)\lambda_0$ because one reflection reverses phase

B) $2nt=m\lambda_0$ because both reflections reverse phase

C) $t=m\lambda_0$ regardless of n

D) Every thickness gives destructive reflection

Q59

+4 / -1

Nonrelativistic electrons accelerated from rest through V have wavelength λ. What wavelength results from acceleration through 4V?

A) 4λ

B) 2λ

C) λ/2

D) λ/4

Q60

+4 / -1

An ideal half-wave rectifier sends $v(t)=V_0 \sin\omega t$ to a resistor R during the positive half-cycle and zero otherwise. What is average resistor power over a full cycle?

A) V_0^2/R

B) $V_0^2/(2R)$

C) $V_0^2/(4R)$

D) $V_0^2/(8R)$

Answer key

One correct option per question. Question and option order are fixed.

Biology

Q	Answer
1	**A**
2	**B**
3	**C**
4	**D**
5	**A**
6	**B**
7	**C**
8	**A**
9	**B**
10	**C**
11	**D**
12	**A**
13	**B**
14	**C**
15	**D**

Chemistry

Q	Answer
16	**A**
17	**B**
18	**D**
19	**A**
20	**B**
21	**C**
22	**A**
23	**B**
24	**D**
25	**C**
26	**B**
27	**D**
28	**D**
29	**A**
30	**C**

Mathematics

Q	Answer
31	**A**
32	**B**
33	**C**
34	**D**
35	**A**
36	**C**
37	**D**
38	**B**
39	**C**
40	**D**
41	**A**
42	**C**
43	**D**
44	**A**
45	**B**

Physics

Q	Answer
46	**A**
47	**B**
48	**D**
49	**C**
50	**D**
51	**B**
52	**A**
53	**D**
54	**B**
55	**C**
56	**D**
57	**A**
58	**B**
59	**C**
60	**C**

Reviewer checklist

Accuracy and single correct answer · Syllabus fit · Distractor quality
Diagram labels · Solution clarity · Difficulty and timed attempt

Reviewer: _____ Date: _____

Decision / notes: _____

Editorial difficulty mix: 9 easy / 42 medium / 9 hard. These labels require human review and are not official item statistics.

Biology

Method, reasoning and common errors

Q1

Biology · medium

Diversity In the Living World / Exposed ovules in gymnosperms

Answer: (A) Ovules are not enclosed in an ovary before fertilisation.

Why: Naked ovules on cone scales identify the gymnosperm condition. A subsequent seed coat does not turn the cone into a fruit, which requires an ovary wall.

Why not the others: (B) Pollen, (C) embryos and (D) vascular tissue occur in both seed-plant groups, so each fails the comparison.

Key point: Compare the ovule's enclosure at pollination, not merely the presence of seeds.

Q2

Biology · hard

Structural Organisation In Plants And Animals / Floral whorls and ovary position

Answer: (B) Inferior, bilocular ovary.

Why: The insertion level of other whorls is above the ovary, defining an epigynous flower with an inferior ovary; a septum divides it into two locules. Rotating a drawing changes its page orientation, not these structural relationships.

Why not the others: (A) and (C) put the attachment below the ovary; (A) and (D) ignore the septum.

Key point: Infer position from attachment, and locule count from internal partitions.

Q3

Biology · medium

Cell: Structure And Functions / Polypeptide hydrolysis and peptide bonds

Answer: (C) 17.

Given: The unbranched chain contains 18 residues.

Formula/Principle: A linear chain of n residues contains $n-1$ peptide linkages; hydrolysis consumes one water per linkage.

Step 1: Count $18-1=17$ bonds.

Step 2: Breaking each bond uses one H_2O, giving 17 molecules.

Why: The terminal residues do not require a bond beyond the chain ends.

Why not the others: (A) counts residues rather than links; (B) subtracts twice; (D) adds a nonexistent terminal bond.

Key point: Distinguish monomer number from linkage number.

Q4

Biology · medium

Plant Physiology / Respiratory quotient and substrate mixture

Answer: (D) 0.8, fat.

Given: CO_2 evolved=16 mL and O_2 consumed=20 mL.

Formula/Principle: RQ=volume CO_2/volume O_2 under comparable conditions.

Step 1: RQ=16/20=0.8.

Step 2: Distance to 1.0 is 0.2, while distance to 0.7 is 0.1, so this reading is closer to the fat reference.

Why: The observed ratio may reflect a mixture of substrates; closeness alone does not prove the exclusive substrate.

Why not the others: (A) and (B) invert the ratio; (C) misjudges the closeness.

Key point: Compute the ratio before comparing reference values.

Q5

Biology · medium

Human Physiology / Primary versus secondary immune response

Answer: (A) Long-lived memory B cells selected during the first exposure.

Why: Priming selects and expands antigen-specific clones. Memory cells respond rapidly on re-exposure and differentiate into antibody-secreting cells, explaining both shorter lag and greater magnitude.

Why not the others: (B) Erythrocytes do not mediate this clonal response; (C) platelets are not antibody producers; (D) neutrophils do not become B-lineage plasma cells.

Key point: The booster changes the available antigen-specific cell population.

Q6

Biology · medium

Reproduction / Double fertilisation and seed tissues

Answer: (B) Seed with embryo and seed coat.

Why: The ovule gives rise to the seed, its integuments form the seed coat, and the zygote forms the embryo. Removal of the ovary wall prevents normal pericarp development, not the ovule's identity.

Why not the others: (A) and (C) describe the ovary-derived fruit wall; (D) is a floral axis, not an ovule derivative.

Key point: Trace seed and fruit structures to different floral precursors.

Q7

Biology · hard

Genetics And Evolution / Selection versus drift in replicate populations

Answer: (C) The data support selection favouring the allele under insecticide.

Why: Parallel directional change across independently maintained large populations under one exposure, together with a no-insecticide comparison, supports differential survival or reproduction. It does not alone prove the molecular mechanism.

Why not the others: (A) Drift is nondirectional across replicates; (B) exposure does not establish directed editing; (D) allele-frequency change does not identify dominance.

Key point: Replication and a control distinguish a selective pattern from chance fluctuations.

Q8

Biology · medium

Biology In Human Welfare / Plasmodium stages and host transfer

Answer: (A) P, sporozoites.

Why: Infected female Anopheles mosquitoes inject salivary-gland sporozoites during feeding. These first reach liver cells, later producing stages that infect erythrocytes; gametocytes are taken up by a subsequent mosquito.

Why not the others: (B) is an intracellular human stage; (C) follows hepatic multiplication; (D) is the sexual precursor ingested from human blood.

Key point: Follow the arrow across hosts before identifying the stage.

Q9

Biology · easy

Biotechnology / Recombinant colony selection

Answer: (B) A recombinant plasmid disrupted tetR.

Why: Growth on ampicillin indicates functional ampR and plasmid uptake; failure on tetracycline indicates loss of tetR activity in the insert-containing construct. The pair of observations provides both positive and negative selection evidence.

Why not the others: (A) predicts ampicillin sensitivity; (C) predicts growth on both drugs; (D) predicts loss of ampicillin growth.

Key point: Interpret the two selective plates together.

Q10

Biology · medium

Ecology / Trophic transfer in a simple food chain

Answer: (C) 80 kJ.

Given: Starting energy=80,000 kJ, with three transfers at 10% each.

Formula/Principle: Available energy after n transfers is initial energy×(0.10)^n.

Step 1: Herbivores receive 8,000 kJ and secondary consumers 800 kJ.

Step 2: Tertiary consumers receive 80 kJ.

Why: The producer-to-tertiary path contains three, not two, trophic transfers.

Why not the others: (A) stops at herbivores; (B) at secondary consumers; (D) adds a fourth transfer.

Key point: Count arrows between trophic levels.

Q11

Biology · medium

Genetics And Evolution / Independent assortment control

Answer: (D) Approximate independent assortment of the two loci.

Why: A double heterozygote crossed to a double recessive reveals its gamete classes directly. Four nearly equal classes fit the 1:1:1:1 expectation; sampling variation explains small deviations.

Why not the others: (A) predicts mainly two parental classes; (B) concerns a dihybrid self, not a testcross; (C) has no support from the four equal counts.

Key point: Match expected ratios to the actual cross design.

Q12

Biology · hard

Genetics And Evolution / Autosomal recessive pedigree conditional risk

Answer: (A) 2/3.

Given: Both unaffected parents must be Aa because they produced aa. The child's genotype probabilities are AA=1/4, Aa=1/2, aa=1/4.

Formula/Principle: Conditional probability divides carrier probability by probability of being unaffected.

Step 1: P(unaffected)=3/4.

Step 2: P(Aa | unaffected)=(1/2)/(3/4)=2/3.

Why: The known unaffected status excludes aa before normalising.

Why not the others: (B) is unconditional carrier probability; (C) is homozygote probability; (D) is probability of being unaffected.

Key point: Update the denominator when a phenotype is observed.

Q13

Biology · medium

Human Physiology / Cardiac output and stroke volume

Answer: (B) Increase by 0.6 L/min.

Given: Initial HR×SV=60×80 mL/min; final=90×60 mL/min.

Formula/Principle: Cardiac output equals heart rate multiplied by stroke volume.

Step 1: Initial output=4,800 mL/min or 4.8 L/min.

Step 2: Final output=5,400 mL/min or 5.4 L/min; difference=+0.6 L/min.

Why: A fall in stroke volume does not guarantee a fall in total output when rate rises enough.

Why not the others: (A) reverses the sign; (C) uses heart-rate change without the volume change; (D) assumes exact compensation.

Key point: Multiply both changing factors before comparing.

Q14

Biology · easy

Cell: Structure And Functions / Osmosis versus facilitated diffusion

Answer: (C) Net water movement follows a water-potential gradient.

Why: Aquaporins provide a selective path for passive water transport; direction depends on the water-potential difference across the membrane. A channel can increase the rate without reversing the thermodynamic direction.

Why not the others: (A) reverses the usual osmotic direction; (B) assigns a pump function to a water channel; (D) incorrectly requires ATP for passive flow.

Key point: Separate transport pathway from driving force.

Q15

Biology · medium

Ecology / Species-area logarithmic slope

Answer: (D) 1/2.

Given: S(4A)/S(A)=40/20=2.

Formula/Principle: In S=cA^z, the ratio for areas 4A and A is 4^z; c cancels.

Step 1: 4^z=2.

Step 2: Because 4^(1/2)=2, z=1/2.

Why: The exponent measures a log–log slope, not the raw species-count change.

Why not the others: (A) predicts a fourfold rise; (B) sixteenfold; (C) only √2-fold.

Key point: Divide paired measurements to eliminate the unknown coefficient.

Chemistry

Method, reasoning and common errors

Q16

Chemistry · easy

Chemical Bonding and Molecular Structure / Hybridisation and sigma-pi count

Answer: (A) 6 σ and 3 π.

Why: Three C–H links and the C–C, C–C and C–N connections each supply one σ bond, totalling six. The double bond adds one π and the triple bond adds two π.

Why not the others: (B) omits one C–H σ; (C) omits a triple-bond π; (D) misallocates one σ in place of π.

Key point: Each bonded atom pair has exactly one σ bond, regardless of bond order.

Q17

Chemistry · medium

Coordination compounds / Oxidation state from complex charge

Answer: (B) +3, 6.

Given: Four neutral ammonia and two −1 chloride ligands give net charge +1.

Formula/Principle: Oxidation state plus ligand charges equals complex charge; coordination number counts donor atoms.

Step 1: $x-2=+1$, hence $x=+3$.

Step 2: Six monodentate ligands supply six donor atoms.

Why: Formal charge and donor count answer different parts of the question.

Why not the others: (A) ignores chloride charge; (C) and (D) count only ammonia donors.

Key point: Count all ligands but include charge only for charged ones.

Q18

Chemistry · medium

Chemical Bonding and Molecular Structure / Formal charge in nitrate resonance

Answer: (D) N +1; each O −1.

Why: N has four bond-order units and no lone pair, assigning formal +1; a singly bonded O has three lone pairs and formal −1. The doubly bonded O is neutral; the total is −1.

Why not the others: (A) gives net −2; (B) net +1; (C) miscounts N's bonding electrons.

Key point: Check the sum of atom formal charges against ion charge.

Q19

Chemistry · hard

Structure of the Atom / Hydrogen emission and orbital occupancy

Answer: (A) 4→2→1.

Given: Photon wavenumber for $n_i \to n_f$ is $R(1/n_f^2 - 1/n_i^2)$.

Formula/Principle: Emission lowers the principal quantum number; the photon energy equals the difference of inverse squares.

Step 1: $R(1/4 - 1/16)$ corresponds to 4→2.

Step 2: $R(1 - 1/4)$ corresponds to 2→1, consistent with the intermediate state.

Why: A proposed sequence must match both energies and the shared intermediate level.

Why not the others: (B) gives $1/4 - 1/9$ first; (C) ends its first transition at 3; (D) gives $1/4 - 1/25$ first.

Key point: Use both spectral lines as a linked path.

Q20

Chemistry · medium

Organic Chemistry – Some Basic Principles and Techniques / Carbocation stability and rearrangement

Answer: (B) The rearrangement can be favourable because hyperconjugation stabilises the tertiary centre.

Why: The migrating H moves with its electron pair from an adjacent carbon; the cation centre shifts to the carbon that lost H. Greater alkyl substitution commonly stabilises a carbocation.

Why not the others: (A) reverses the stated stability trend; (C) invokes an external source; (D) violates charge conservation.

Key point: Follow the electron pair and compare the two carbocation environments.

Q21

Chemistry · easy

Aldehydes, ketones, and carboxylic acids / Acyl derivative hydrolysis

Answer: (C) CH_3COOH and HCl.

Why: Water substitutes the acid chloride's Cl at the acyl carbon to give acetic acid; the leaving chloride combines with a proton to give HCl. Carbon skeleton and oxidation state remain those of the acid derivative.

Why not the others: (A) invents an aldehyde and chlorine; (B) breaks the C–C framework; (D) forms a ketone rather than hydrolysing the acyl chloride.

Key point: Hydrolysis replaces the acyl leaving group with OH.

Q22

Chemistry · medium

Organic compounds containing nitrogen / Hinsberg test on amine classes

Answer: (A) Primary amine.

Why: A primary amine yields a sulfonamide retaining N–H; base removes that proton to make a water-soluble salt. Acidification regenerates the less soluble neutral sulfonamide.

Why not the others: (B) Secondary-amine sulfonamide lacks N–H and does not show this base solubility; (C) cannot form the same sulfonamide; (D) is permanently charged and not an amine of this class.

Key point: The acid–base behaviour tests whether sulfonamide N–H remains.

Q23

Chemistry · medium

Biomolecules / Amino acid charge and isoelectric point

Answer: (B) H_3N^+–CHR–COO^-, net 0.

Why: pH 6 lies above the carboxyl pK_a, so carboxyl is largely deprotonated, but below the ammonium pK_a, so nitrogen remains largely protonated. Opposite unit charges cancel.

Why not the others: (A) requires more acidic pH; (C) more basic pH; (D) combines protonation states inconsistent with both pK_a values.

Key point: Compare pH separately with each ionisable group's pK_a.

Q24

Chemistry · medium

Chemical Kinetics / Arrhenius plot and activation energy

Answer: (D) Y, by a factor of two.

Why: Arrhenius linearisation gives slope $-E_a/R$ for ln k versus 1/T. Y's slope magnitude is twice X's, so its barrier is twice as large when the same R applies. Intercepts affect pre-exponential factors, not the slope comparison.

Why not the others: (A) reverses the slopes; (B) ignores their magnitudes; (C) would apply only without the axis and Arrhenius assumptions.

Key point: Use absolute slope, since the minus sign represents temperature dependence.

Q25

Chemistry · medium

Equilibrium / Common-ion solubility approximation

Answer: (C) 10^{-8} M.

Given: Ksp=$[Ag^+][Cl^-]=10^{-10}$ and $[Cl^-]\approx10^{-2}$ M.

Formula/Principle: AgCl solubility s equals $[Ag^+]$ at equilibrium.

Step 1: $s\times10^{-2}\approx10^{-10}$.

Step 2: $s\approx10^{-8}$ M, far below 0.010 M, confirming the approximation.

Why: Added common ion suppresses dissolution.

Why not the others: (A) and (B) understate chloride suppression; (D) divides by chloride twice.

Key point: Check the approximation after solving.

Q26

Chemistry · hard

Redox Reactions / Disproportionation of chlorine in alkali

Answer: (B) One chlorine atom loses one electron while another gains one electron.

Why: Cl starts at 0 in Cl_2; one product chlorine is −1 in chloride, the other is +1 in hypochlorite. The paired one-electron reduction and oxidation balance internally in disproportionation.

Why not the others: (A) neglects hypochlorite; (C) averages −1 and +1 incorrectly; (D) leaves oxygen at −2.

Key point: Assign oxidation states to each chlorine product separately.

Q27

Chemistry · medium

Thermodynamics / Calorimetric final temperature

Answer: (D) 50 °C.

Given: Both samples have equal mass and specific heat.

Formula/Principle: Heat lost by hot sample equals heat gained by cold sample.

Step 1: $mc(80-T)=mc(T-20)$.

Step 2: $80-T=T-20$, giving T=50 °C.

Why: Equal thermal capacities make the arithmetic midpoint the energy-balance result.

Why not the others: (A) or (B) would require zero change in one sample; (C) gives unequal heat magnitudes.

Key point: The midpoint works only when the heat capacities match.

Q28

Chemistry · medium

Solutions / Raoult partial pressure from mixture

Answer: (D) 2/5.

Given: xA=.25 and xB=.75.

Formula/Principle: Raoult's law gives $p_i = x_i p_i^\circ$.

Step 1: pA=.25×80=20 kPa; pB=.75×40=30 kPa.

Step 2: Fraction pA/(pA+pB)=20/50=2/5.

Why: Vapour composition differs from liquid composition because pure-component pressures differ.

Why not the others: (A) assumes equal partial pressures; (B) and (C) use an incorrect total or exchange the two vapour shares.

Key point: Calculate both partial pressures before taking a share.

Q29

Chemistry · medium

Hydrocarbons / Electrophilic aromatic substitution directing effects

Answer: (A) Replace CH_3 with NO_2.

Why: Nitro withdraws electron density by resonance and induction, destabilising ortho/para sigma complexes relative to meta attack. Methyl donation instead favours ortho/para substitution.

Why not the others: (B) and (C) donate by resonance and direct ortho/para; (D) is deactivating but still ortho/para directing by lone-pair resonance.

Key point: Direction and overall reaction rate are separate substituent effects.

Q30

Chemistry · medium

Haloalkanes and haloarenes / Aryl versus alkyl nucleophilic substitution

Answer: (C) Chlorobenzene's aryl C–Cl bond has partial double-bond character and its carbon is sp^2.

Why: Lone-pair conjugation strengthens the aryl bond, ordinary SN2 approach at sp^2 carbon is disfavoured, and phenyl cation formation is costly. Benzylic carbon is sp^3 and its transition states/intermediates can be resonance stabilised.

Why not the others: (A) is false; (B) reverses cation stability; (D) falsely excludes benzylic SN2.

Key point: Compare the carbon bearing chlorine, not simply whether a ring is present.

Mathematics

Method, reasoning and common errors

Q31

Mathematics · medium

Coordinate Geometry / Hyperbola asymptotes from vertices and focus

Answer: (A) $y=\pm(4/3)x$.

Given: Horizontal transverse axis, $a=3$ and $c=5$.

Formula/Principle: For $x^2/a^2-y^2/b^2=1$, $c^2=a^2+b^2$ and asymptotes are $y=\pm(b/a)x$.

Step 1: $b^2=25-9=16$, so $b=4$.

Step 2: Slopes are $\pm4/3$.

Why: The focal distance is not itself the conjugate semiaxis.

Why not the others: (B) swaps a and b; (C) uses c in place of b; (D) mixes b with c.

Key point: Recover the missing semiaxis before finding slope.

Q32

Mathematics · medium

Coordinate Geometry / Common point of a line family

Answer: (B) (7/3,1/3).

Given: The point must satisfy the equation for every real k.

Formula/Principle: Equate separately coefficients of k and constant terms in an identity.

Step 1: k terms give $x+2y=3$; constants give $x-y=2$.

Step 2: $y=1/3$, $x=7/3$.

Why: A fixed point must survive any choice of the line parameter, not just one selected k.

Why not the others: (A) misses both equations; (C) satisfies $x-y=2$ but not $x+2y=3$; (D) misses the constant equation.

Key point: Separate parameter-dependent and independent parts.

Q33

Mathematics · hard

Integration / Definite integral of rational function

Answer: (C) ln 3.

Given: The denominator factors as (x+1)(x+2).

Formula/Principle: Use partial fractions and evaluate at both bounds.

Step 1: $(2x+3)/[(x+1)(x+2)]=1/(x+1)+1/(x+2)$.

Step 2: $[\ln(x+1)+\ln(x+2)]_0^1=\ln2+\ln3-\ln2=\ln3$.

Why: Both logarithms contribute boundary terms; the ln2 at the upper bound cancels the ln2 at the lower bound of the other term.

Why not the others: (A) omits a surviving log; (B) integrates only 1/(x+2); (D) omits the lower endpoint.

Key point: Simplify evaluated logarithms before selecting a result.

Q34

Mathematics · medium

Integration / Area between parabola and line

Answer: (D) 4/3.

Given: The curves meet at x=0 and x=2.

Formula/Principle: Area is ∫(upper−lower) dx, never a signed cancellation.

Step 1: On (0,2), $2x$ exceeds x^2.

Step 2: $\int_0^2(2x-x^2)dx=[x^2-x^3/3]_0^2=4-8/3=4/3$.

Why: The linear curve stays above until the second intersection.

Why not the others: (A) and (B) lose a factor from the interval; (C) drops the subtraction.

Key point: Determine intersections and upper curve first.

Q35

Mathematics · medium

Differentiation / Implicit tangent of a level curve

Answer: (A) −4/5.

Given: x^2+xy+y^2 is constant along the curve.

Formula/Principle: Differentiate implicitly, treating y as a function of x.

Step 1: $2x+y+xy'+2yy'=0$, so $y'=-(2x+y)/(x+2y)$.

Step 2: At (1,2), $y'=-4/5$.

Why: The product xy contributes both y and xy'.

Why not the others: (B) inverts the derivative; (C) misses the sign; (D) drops the x dependence.

Key point: Apply the product rule to xy.

Q36

Mathematics · medium

Relations & functions / Equivalence relation modulo a divisor

Answer: (C) 2.

Why: Congruence modulo 4 is reflexive, symmetric and transitive, so restriction to a subset preserves those properties. Odd integers fall into residues 1 or 3 modulo 4, both nonempty and distinct.

Why not the others: (A) merges residues 1 and 3; (B) includes even residue classes absent from the domain; (D) mistakes infinitely many members for infinitely many classes.

Key point: Count equivalence classes on the stated domain, not all integers.

Q37

Mathematics · medium

Matrices & determinants / Matrix equation and noncommutativity

Answer: (D) 1.

Given: $AX=B$, so $X=A^{-1}B$, not BA^{-1}.

Formula/Principle: $A^{-1}=[[1,-1],[0,1]]$.

Step 1: $A^{-1}B=[[0,-1],[1,1]]$.

Step 2: Its trace is $0+1=1$.

Why: Left multiplication preserves the order dictated by $AX=B$; matrix multiplication is not commutative.

Why not the others: (A) adds off-diagonal entries; (B) uses the diagonal of B; (C) omits the lower-right entry.

Key point: Solve the matrix equation on the correct side.

Q38

Mathematics · hard

Probability / Conditional binomial event

Answer: (B) 2/5.

Given: 16 equally likely sequences; one has no heads.

Formula/Principle: P(exactly two | at least one)=P(exactly two)/P(at least one).

Step 1: Six sequences have two heads, while 15 satisfy the condition.

Step 2: Required probability=6/15=2/5.

Why: The conditioning event changes the denominator, not the six favourable sequences.

Why not the others: (A) and (D) retain 16 as denominator; (C) counts an incorrect number of favourable cases.

Key point: Restrict the sample space before forming a ratio.

Q39

Mathematics · medium

Trigonometric functions / Compound-angle cosine under quadrant constraints

Answer: (C) $1/\sqrt{2}$.

Given: Acute angles give positive sine and cosine.

Formula/Principle: $\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$.

Step 1: $(\cos\alpha,\sin\alpha)=(2,1)/\sqrt{5}$ and $(\cos\beta,\sin\beta)=(3,1)/\sqrt{10}$.

Step 2: $(6-1)/\sqrt{50}=5/\sqrt{50}=1/\sqrt{2}$.

Why: Quadrant information fixes the signs, while the subtraction reflects the compound angle.

Why not the others: (A) and (B) omit a term; (D) ignores β.

Key point: Build unit-triangle components before combining angles.

Q40

Mathematics · easy

Three-dimensional geometry / Foot on coordinate plane

Answer: (D) $(2,-3,0)$.

Given: The xy-plane consists of $z=0$.

Formula/Principle: An orthogonal projection onto that plane changes only the normal (z) coordinate.

Step 1: Keep $x=2$ and $y=-3$.

Step 2: Set $z=0$.

Why: The connecting segment is parallel to the z-axis and perpendicular to the plane.

Why not the others: (A) and (B) change a coordinate within the plane; (C) changes both in-plane coordinates.

Key point: Identify the plane's normal direction.

Q41

Mathematics · medium

Vectors / Cross product from area and orientation

Answer: (A) $(0,0,-3)$.

Given: Both vectors lie in the xy-plane.

Formula/Principle: The z-component of $a\times b$ is $a_x b_y-a_y b_x$.

Step 1: $1\times1-2\times2=-3$.

Step 2: The other components vanish, giving $-3\hat{k}$.

Why: Magnitude 3 is area, but order $a\times b$ sets the negative orientation.

Why not the others: (B) reverses order; (C) is in-plane; (D) uses the dot-product sum.

Key point: Preserve operand order in cross products.

Q42

Mathematics · medium

Complex Numbers / Cube roots of unity as geometric points

Answer: (C) 3.

Given: $\omega=-1/2\pm i\sqrt{3}/2$, a vertex of the unit-circle equilateral triangle.

Formula/Principle: $|z|^2=z\,\bar{z}$, or sum of squared real and imaginary parts.

Step 1: $1-\omega=3/2\mp i\sqrt{3}/2$.

Step 2: Modulus squared $=9/4+3/4=3$.

Why: Both non-real conjugate choices have the same distance from 1.

Why not the others: (A) uses $|\omega|^2$; (B) omits the real shift; (D) would correspond to opposite unit-circle points.

Key point: Geometric distance is invariant under conjugation here.

Q43

Mathematics · medium

Quadratic Equations / Two positive real roots

Answer: (D) $t>4$.

Given: Product of roots $=4>0$ and sum $=t$.

Formula/Principle: Distinct real roots require discriminant $t^2-16>0$; positive sum selects positive rather than negative roots.

Step 1: Real distinct gives $t<-4$ or $t>4$.

Step 2: Positive sum removes $t<-4$, leaving $t>4$.

Why: At $t=4$ the two positive roots coincide, violating distinctness.

Why not the others: (A) admits non-real roots; (B) includes the double root; (C) gives two negative roots.

Key point: Combine discriminant and Vieta signs with strict endpoints.

Q44

Mathematics · medium

Binomial Theorem / Optimising a constant-term coefficient across exponents

Answer: (A) 2.

Given: Choosing r factors of $x^{(-p)}$ gives power $2(6-r)-pr=12-(p+2)r$.

Formula/Principle: A constant term needs integer $r=12/(p+2)$ between 0 and 6; its coefficient is $C(6,r)$.

Step 1: For $p=1,2,4,10$, the allowed r values are 4,3,2,1.

Step 2: Corresponding coefficients are 15,20,15,6, so $p=2$ gives the maximum.

Why: Exponent cancellation alone does not determine which of several valid expansions has the largest coefficient.

Why not the others: (B) and (C) each give 15; (D) gives 6.

Key point: Compare the binomial coefficients only after finding each admissible constant term.

Q45

Mathematics · easy

Sequences & series / Geometric-series convergence boundary

Answer: (B) $|r|<1$.

Why: The nth partial sum is $(1-r^n)/(1-r)$ for $r\neq1$. It approaches $1/(1-r)$ only when $r^n\to0$, precisely for $-1<r<1$; $r=1$ grows and $r=-1$ oscillates.

Why not the others: (A) and (D) each admit ratios of magnitude at least one; (C) incorrectly includes both boundary values.

Key point: Test endpoints rather than silently extending a finite-series formula.

Physics

Method, reasoning and common errors

Q46

Physics · medium

Units and Measurements / Maximum fractional uncertainty in resistance

Answer: (A) 3%.

Given: Fractional maximum bounds are 0.02 in V and 0.01 in I.

Formula/Principle: First-order worst-case relative errors add for a quotient.

Step 1: $\delta R/R \approx \delta V/V - \delta I/I$ as signed perturbations.

Step 2: Maximum magnitude $\leq 2\% + 1\% = 3\%$.

Why: Opposite signed errors can reinforce in the quotient; subtracting percentages is not a safe upper bound.

Why not the others: (B) ignores current; (C) ignores voltage; (D) multiplies percentage numbers.

Key point: Distinguish worst-case uncertainty from a specific signed deviation.

Q47

Physics · easy

Laws of Motion / Static friction at threshold

Answer: (B) 20 N.

Given: Normal reaction $= mg = 50$ N on the horizontal floor.

Formula/Principle: Limiting static friction $f_{s,max} = \mu_s N$.

Step 1: $f_{s,max} = .4 \times 50 = 20$ N.

Step 2: At the threshold the applied force balances this maximum.

Why: Static friction adjusts to smaller forces before its limit is reached.

Why not the others: (A) halves the result; (C) uses N itself; (D) omits the mass factor.

Key point: Use the limiting value only at impending motion.

Q48

Physics · medium

Mechanical Properties of Solids / Series extension of equal wires

Answer: (D) 3FL/(2AY).

Given: Both series wires carry the same tension F.

Formula/Principle: Each extension equals FL/(AY_i); series extensions add.

Step 1: First wire extends FL/(AY), second FL/(2AY).

Step 2: Sum=3FL/(2AY).

Why: Equal force does not imply equal strain for different moduli.

Why not the others: (A) retains only the stiffer wire; (B) only the first; (C) assumes identical moduli.

Key point: Add displacements for elements in series.

Q49

Physics · medium

Thermal Properties of Matter / Differential thermal expansion in a ring-gap arrangement

Answer: (C) The sphere can no longer pass through the hole.

Why: A hole expands as if it were made of the plate material; its diameter scales by approximately $1+\alpha\Delta T$, while the sphere diameter scales by $1+2\alpha\Delta T$. Starting equal, the sphere becomes larger.

Why not the others: (A) falsely makes a heated hole contract; (B) ignores the specified coefficients; (D) reverses their ratio.

Key point: Apply each material's linear expansion to its own diameter.

Q50

Physics · hard

Kinetic Theory / Equipartition and internal energy

Answer: (D) $2T_0$.

Given: Diatomic gas has five active quadratic degrees of freedom, n=1, and work on gas=$5RT_0/2$.

Formula/Principle: $U=(5/2)nRT$ and first law $\Delta U=Q+W_on$.

Step 1: $Q=0$ and $\Delta U=(5/2)R(T_f-T_0)=5RT_0/2$.

Step 2: $T_f-T_0=T_0$, hence $T_f=2T_0$.

Why: Mechanical stirring raises internal energy despite the container being thermally insulated.

Why not the others: (A) halves the temperature rise; (B) mistakes work in units RT for final T; (C) doubles the rise.

Key point: Insulated means Q=0, not $\Delta U=0$.

Q51

Physics · medium

Oscillations / Pendulum period under length change

Answer: (B) 1.20.

Given: Lnew/Lold=1.44.

Formula/Principle: T=2π√(L/g) for a simple pendulum at small amplitude.

Step 1: Tnew/Told=√1.44.

Step 2: Ratio=1.20.

Why: Period varies with square root of length rather than length itself.

Why not the others: (A) uses direct proportionality; (C) takes reciprocal; (D) squares length ratio.

Key point: Cancel common factors before taking the square root.

Q52

Physics · easy

Waves / Beat-frequency ambiguity with two references

Answer: (A) 252 Hz.

Given: |f−256|=4 and |f−260|=8.

Formula/Principle: Beat frequency is the absolute difference of two close frequencies.

Step 1: First reference permits f=252 or 260 Hz.

Step 2: Against 260 Hz, these give 8 or 0 beats/s; only 252 qualifies.

Why: One beat count alone has a high/low ambiguity.

Why not the others: (B) gives 8 beats against 256; (C) gives none against 260; (D) gives none against 256.

Key point: Intersect candidate sets from two reference tones.

Q53

Physics · medium

Electric Charges and Fields / Electrostatic Potential / Potential difference across a uniform field

Answer: (D) 4 V.

Given: E_x=+3 V/m over 2 m.

Formula/Principle: $\Delta V = -\int E_x\, dx$.

Step 1: V(2)−V(0)=−3×2=−6 V.

Step 2: V(2)=10−6=4 V.

Why: Electric potential decreases along the field direction for positive E.

Why not the others: (A) reverses sign; (B) ignores field; (C) uses only one metre.

Key point: Keep the minus sign between field and potential gradient.

Q54

Physics · medium

Current Electricity / Drift speed and current density

Answer: (B) J.

Given: $n' = n/2$ and $v' = 2v$.

Formula/Principle: Drift current-density magnitude $J = nqv$.

Step 1: $J' = (n/2)q(2v)$.

Step 2: Factors cancel, so $J' = J$.

Why: Fewer carriers can yield the same flow per area if each drifts proportionally faster.

Why not the others: (A) divides twice; (C) changes only v; (D) compounds two increases that are not both present.

Key point: Current density depends on the product of density and drift speed.

Q55

Physics · hard

Electromagnetic Induction / Inductor energy and current scaling

Answer: (C) 2U.

Given: Final $L' = L/2$ and $I' = 2I$.

Formula/Principle: Magnetic energy $U = (1/2)LI^2$.

Step 1: Doubling I alone multiplies energy by four.

Step 2: Halving L at that current halves 4U to 2U.

Why: Energy depends quadratically on current but linearly on inductance; compare the final state with the initial one.

Why not the others: (A) changes only L; (B) treats effects as cancelling linearly; (D) ignores the L change.

Key point: Apply both changes to the same energy expression.

Q56

Physics · medium

Alternating Current / Resonance curve and bandwidth

Answer: (D) P has smaller resistance and smaller bandwidth.

Why: At series resonance impedance is R, so peak current V/R is larger for smaller R. Bandwidth in angular frequency is R/L, so the same smaller R narrows the peak with L fixed.

Why not the others: (A) and (B) predict a shorter peak for greater R; (C) conflicts with the common L and C setting the same ideal resonance.

Key point: Read both peak height and width, not one feature alone.

Q57

Physics · easy

Ray Optics and Optical Instruments / Prism minimum deviation and refractive index

Answer: (A) √2.

Given: Prism angle A=60°, minimum deviation δm=30°.

Formula/Principle: n=sin[(A+δm)/2]/sin(A/2).

Step 1: Numerator sin45°=1/√2 and denominator sin30°=1/2.

Step 2: Ratio=√2.

Why: At minimum deviation the two refractions are symmetric, giving the stated formula.

Why not the others: (B) uses an incorrect refraction angle; (C) neglects sin45°; (D) inverts the ratio.

Key point: Half both the sum and the prism angle.

Q58

Physics · medium

Wave Optics / Thin-film reflected phase reversals

Answer: (B) $2nt=m\lambda_0$ because both reflections reverse phase.

Why: Reflection from air to film and from film to higher-index glass each adds a π phase reversal, so their relative reflection phase is zero. The remaining round-trip optical path is 2nt and must equal an integer free-space wavelength.

Why not the others: (A) assumes only one reversal; (C) omits the round trip and index; (D) ignores path-dependent phase.

Key point: Count relative, not absolute, reflection phase changes.

Q59

Physics · medium

Dual Nature of Radiation and Matter / Electron de Broglie wavelength under acceleration

Answer: (C) λ/2.

Given: Electron charge and mass are unchanged; kinetic energy eV.

Formula/Principle: λ=h/√(2meV) in the nonrelativistic limit.

Step 1: Voltage grows by factor four, so momentum grows by √4=2.

Step 2: Wavelength halves.

Why: Accelerating potential sets kinetic energy, not momentum directly.

Why not the others: (A) and (B) increase λ with voltage; (D) scales λ inversely with V rather than √V.

Key point: Keep the square-root dependence.

Q60

Physics · medium

Semiconductor Electronics / Half-wave rectified mean power

Answer: (C) $V_0^2/(4R)$.

Given: Power=v^2/R for half a cycle and zero for half a cycle.

Formula/Principle: Average $\sin^2$ over a positive half-cycle is 1/2.

Step 1: Mean over that active half-cycle is $V_0^2/(2R)$.

Step 2: Only half the full period is active, yielding $V_0^2/(4R)$.

Why: Mean power uses squared voltage, not the square of average voltage.

Why not the others: (A) uses peak power; (B) averages only the conducting half; (D) introduces an extra factor two.

Key point: Average over the full period.